

Assignment1

1. What is the range of N bits integers with 2's complement (补码) ? What is the range of N bits integers with 1's complement (反码) ? Are they the same? Explain the reasons.

They are not the same.

For N bits integers with 2's complement, its positive range is from $00 \cdots 00_{N \text{ bits}}$ to $01 \cdots 11_{N \text{ bits}}$, which is from 0 to $2^{N-1} - 1$

Its negative range is from $11 \cdots 11_{N \text{ bits}}$ to $10 \cdots 00_{N \text{ bits}}$. where $11 \cdots 11_{N \text{ bits}}$ with 2's complement is negative $0 \cdots 01_{N-1 \text{ bits}}$, which is -1. $10 \cdots 00_{N \text{ bits}}$ with 2's complement is negative $10 \cdots 00_{N \text{ bits}}$, which is -2^{N-1} .

比如N=3, 负数补码111=-1, 负数100=-4, 负数的原码范围是: -4到-1.

As a result, the range of N bits integers with 2's complement is from -2^{N-1} to $2^{N-1} - 1$.

For N bits integers with 1's complement, its positive range is from

$00 \cdots 00_{N \text{ bits}}$ to $01 \cdots 11_{N \text{ bits}}$, which is from 0 to $2^{N-1} - 1$

Its negative range is from $11 \cdots 11_{N \text{ bits}}$ to $10 \cdots 00_{N \text{ bits}}$. where $11 \cdots 11_{N \text{ bits}}$ with 1's complement is negative $0 \cdots 00_{N-1 \text{ bits}}$, which is -0.

$10 \cdots 00_{N \text{ bits}}$ with 1's complement is negative $1 \cdots 11_{N-1 \text{ bits}}$, which is $-2^{N-1} + 1$.

比如N=3, 负数反码111=0, 负数反码100=-3, 负数的原码范围是: -3到0.

As a result, the range of N bits integers with 2's complement is from $-2^{N-1} + 1$ to $2^{N-1} - 1$.

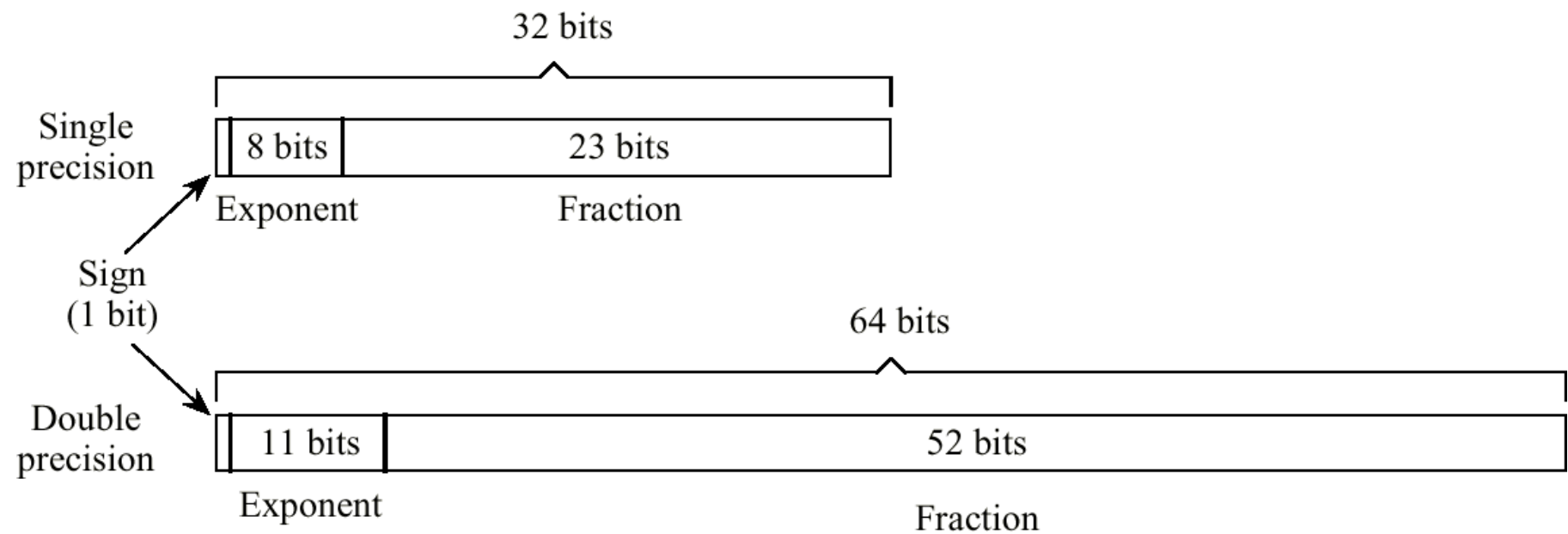
Bit pattern	Value represented
011	3
010	2
001	1
000	0
111	-1
110	-2
101	-3
100	-4

2. Represent the following decimal floating numbers using binary numbers following IEEE754 single precision standard

$$3\frac{3}{8}$$
$$-2\frac{1}{2}$$

Reference: an IEEE754 single precision number is calculated as:

$$(-1)^{sign\ bit} * (1 + mantissa) * 2^{(exponent-127)}$$



$3\frac{3}{8}$
 $-2\frac{1}{2}$

11.011 -> 0 (sign bit) .1011 (mantissa) 1 (exponent :1+127=128->10000000) , ->
0 10000000 101100000000000000000000

-10.1 -> 1 (sign bit) .01 (mantissa) 1 (exponent :1+127=128->10000000) ->
1 10000000 010000000000000000000000

3. Suppose registers E and F contained $(AA)_{\text{hex}}$ and $(CC)_{\text{hex}}$, respectively. What bit pattern would be in register D after executing each of the following instructions (see appendix)?

- A. 7DEF _____ A: $(EE)_{\text{hex}}$
1 Opcode 7 means OR the bit patterns in registers... (see textbook Appendix C page 552)
2 Convert $(AA)_{\text{hex}}$ to binary presentation: 10101010
3 Convert $(CC)_{\text{hex}}$ to binary presentation: 11001100
4 The result of OR operation is 11101110, which is $(EE)_{\text{hex}}$
- B. 8DEF _____ B: $(88)_{\text{hex}}$
1 Opcode 8 means ADD the bit patterns in registers... (see textbook Appendix C page 552)
2 Convert $(AA)_{\text{hex}}$ to binary presentation: 10101010
3 Convert $(CC)_{\text{hex}}$ to binary presentation: 11001100
4 The result of ADD operation is 10001000, which is $(88)_{\text{hex}}$
- C. 9DEF _____ C: $(66)_{\text{hex}}$
1 Opcode 9 means EXCLUSIVE OR the bit patterns in registers... (see textbook Appendix C page 552)
2 Convert $(AA)_{\text{hex}}$ to binary presentation: 10101010
3 Convert $(CC)_{\text{hex}}$ to binary presentation: 11001100
The result of OR operation is 01100110, which is $(66)_{\text{hex}}$

4. High definition video can be delivered at a rate of 30 frames per second (fps) where each frame has a resolution of 1920 1080 pixels using 3Bytes per pixel. Can an uncompressed video stream of this format be sent over a USB 1.1 serial port? USB 2.0 serial port? USB 3.0 serial port? Why? (**Note**: The maximum speeds of USB 1.1, USB 2.0, and USB 3.0 serial ports are 12Mbps, 480Mbps, and 5Gbps respectively.)

The HD video stream take $1920 \times 1080 \times 3 \times 8 \times 30$ bits/second which is about 1.5Gbps.

Be careful about the definition of Mbps and Gbps (referring to page 98).

5. The following is a routine encoded in the machine language described in the language description table. Explain (in a single sentence) what the routine does. (Explain what the entire routine does as a unit rather than reciting what each instruction does.)

210F, 12A0, 8212, 32A0

Opcode 2 means load the register with a bit pattern (see textbook Appendix C page 551). Therefore, 210F means load the register 1 with $(0F)_{\text{hex}}$.

Opcode 1 means load the register with the bit pattern in a memory cell (see textbook Appendix C page 551). Therefore, 12A0 means load the register 2 with the bit pattern stored in memory cell A0.

Opcode 8 means ADD the bit patterns in registers (see textbook Appendix C page 552). Therefore, 8212 means ADD the bit pattern stored in register 1 and 2, and store the result in register 2.

Opcode 3 means STORE the bit patterns to a memory cell (see textbook Appendix C page 552). Therefore, 32A0 means store the bit pattern stored in register 2 into memory cell A0. Of course the exact value is unknown.

6. Suppose the memory cells at addresses 20 through 28 contain the following bit patterns (the program counter contains 20):

Address	Contents
20	12
21	20
22	32
23	30
24	B0
25	21
26	20
27	C0
28	00

(1) What bit patterns will be in register 0, 1 and 2 when the machine halts?

(2) What bit pattern will be in the memory cell at address 30 when the machine halts?

(3) What bit pattern will be in the memory cell at address B0 when the machine halts?

(1) Registers 0, 1 and 2 will contain 32, 20 and 12, respectively.

(2) 12

(3) 32

1220---load the register 2 with the bit pattern found in the memory cell whose address is 20

3230--- Store the bit pattern found in register 2 in the memory cell whose address is 30

B021---Jump to the instruction located in the memory cell at address 21.

2032---load register 0 with 32

30B0---Store the bit pattern found in register 0 in the memory cell whose address is B0

2120--- load register 1 with 20

C000--- halt execution.

7. Using the machine language described in the appendix, write a sequence of instructions that will place a 1 in the most significant bit of the memory cell at address A0 without disturbing the other bits.

First, let's think about how to place a 1 in the most significant bit of a bit pattern. The most significant bit is the left-most bit of the bit pattern (defined within our course). We are not aware of its state (0 or 1), but we only have to assure its value being 1 after the operation. As a result, we adopt masking technique (referring to chapter 2). If we design a mask with its most significant bit being 1, e.g. 1xxxxxx... (x represent 0 or 1), then we employ OR operation to "OR" the bit pattern with the mask. Then the most significant bit will be 1, correct? Let's move on, the question reads: "without disturbing the other bits", which means the previous operation should not change any bit except the most significant one. Finally we see that our mask should be 1000000...

So, the answer is:

(Assume we use 8 bits)

First, 2180 – load register 1 with 10000000.

Second, 12A0 – load register 2 with bit pattern stored in memory cell A0.

Then, 7312 – store register 3 with the result of "OR" 1 and 2.

The last, 33A0 – write the result back.

- 8. Suppose the registers 4 and 5 contain the bit patterns 3C and C8, respectively. What bit pattern is left in register 0 after executing each of the following instructions:
- 5045 a. ADD 3C and C8-----04
- 7045 c. OR 3C and C8-----FC
- 8045 d. AND 3C and C8-----08
- 9045 e. XOR 3C and C8-----F4

9. Suppose the memory cells at addresses AF through B1 in the machine described in the Appendix contain the following bit patterns

<u>Address</u>	<u>Contents</u>
AF	B0
B0	B0
B1	AF

What would happen if we start the machine with its program counter containing AF?

B	RXY	JUMP to the instruction located in the memory cell at address XY if the bit pattern in register R is equal to the bit pattern in register number 0. Otherwise, continue with the normal sequence of execution. (The jump is implemented by copying XY into the program counter during the execute phase.) <i>Example:</i> B43C would first compare the contents of register 4 with the contents of register 0. If the two were equal, the pattern 3C would be placed in the program counter so that the next instruction executed would be the one located at that memory address. Otherwise, nothing would be done and program execution would continue in its normal sequence.
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The machine will alternate between executing the jump instruction at address AF and the jump instruction at address B0